Virtual Vials

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1. Overview

1.1 Introduction

- Virtual Vials will consist mainly of developing a VR-based learning environment for chemistry experiments. It will involve students simulating a laboratory where they could practice handling chemicals and/or working in a lab setting, learning lab protocol and safety in an engaging and effective way.

1.2 Problem

- Nowadays, when learning something new, motivation and sometimes even the cost often become significant barriers to learning. Motivation is dependent on students; some may struggle to stay motivated over time, especially without any clear goal or rewards. Additionally, financial costs such as fees for books and materials can also play a major role in learning.

1.3 Solution

- Virtual Vials offers a solution to the aforementioned problem. The app will offer a blend of learning and engagement with the goal of helping students increase their productivity, as it supplements traditional learning methods. Additionally, learning may feel satisfying as it involves rewards that may be earned in-game, which, as a result, gives a sense of willingness and satisfaction.
- Virtual Vial is a 3D simulation; therefore, the app is easily accessible no matter the location. All students can participate, with any level of experience or background, and gain access to hands-on experience with chemicals and equipment in a “real” environment without running into risks of breaking or messing with real equipment/chemicals. As opposed to 3D learning, which is limited to observations and reading, which reduces the depth of experience. Moreover, VR simulations can provide real-time feedback based on the manipulation and interaction of chemicals and lab equipment, giving the feeling of realism and immersion, which encourages trial-and-error learning and deeper engagement. 3D learning is often indirect and lacks the feeling of being “real,” which makes it harder to build muscle memory linked to specific actions.

2. Game Concept

2.1 Description

- A way for students and trainees alike to handle and work with different chemicals in a remote way. It also lets them learn different safety lab protocols and scenarios that supplement their module readings, which allows for a real-world application of the chemistry lab environment.

2.2 Setting

- The whole simulation will take place in a real-world laboratory setting.

2.3 Audience

- The target audience of the project is mainly high school/university students

3. Gameplay

3.1 Mechanics

- The game will have three different modes:
    - ➔ **Standard:** The main premise of the project, to simulate a lab setting where a trainee can use different equipment in a laboratory to mix and handle different chemicals to see their reaction.
    - ➔ **Timed:** A gamemode where the player has to mix specific different reactions until the timer runs out. Successfully replicating the reaction will grant them more time.
    - ➔ **Safety:** A gamemode where the trainee has to find all the safety problems in the lab.

3.2 Controls

	VR Handheld Controllers
Move Right	Left Joystick Right
Move Left	Left Joystick Left
Move Forward	Left Joystick Up
Move Backward	Left Joystick Back
Turn Right/left	Right Joystick Right/left
Grab	Right Grip
Interact	Left Trigger

- Controls may be reprogrammable further in development, the above are default controls.

3.3 Level Design / Level Mechanics

- As the player in the normal, standard mode, they will progress by successfully completing each experiment/lesson as they progress through the material

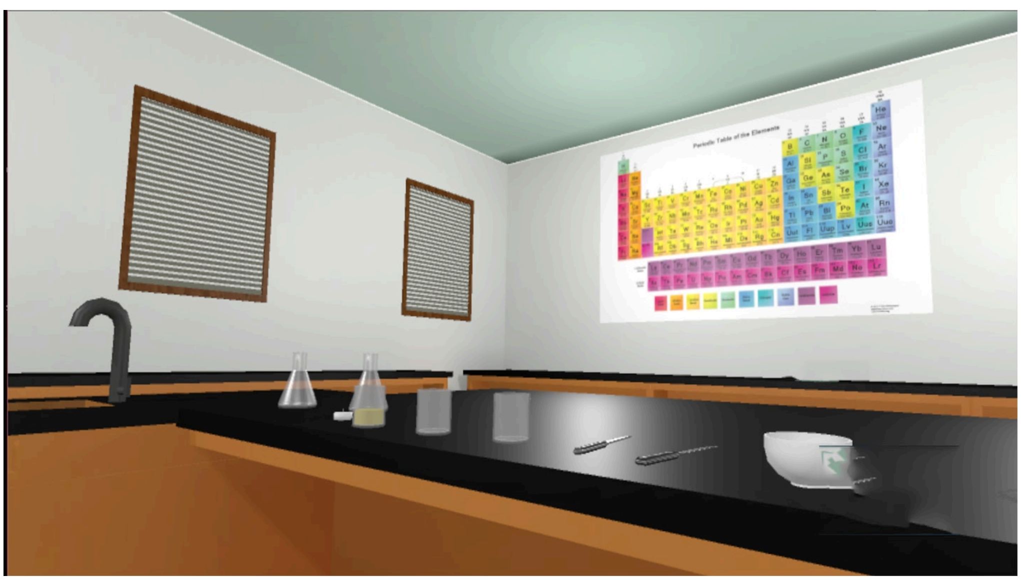
- Completion will be determined on successful completion of the lesson as well as standard lab safety requirements
- Future lessons will build on the previous to lead to a progressive difficulty level

3.4 User Interface

- The general start User interface will include a title screen to select options and the different modes of the game, as well as a quit button.
    - Play
        - Standard
        - Timed
        - Safety
    - Options
    - Quit
- Inside a mode, while playing the game, the ui will consist of pop-up, interactable bubbles explaining the lesson and the steps to complete them as the core ui.

4. Arts and Visuals

- The game will take place in a laboratory with equipment you would expect in a lab environment.



Images from: The VR Chemistry Lab

5. Technical Details/Requirements

5.1 Hardware Requirements

- Virtual Vials will run on both PC and standalone VR headsets. As the game is developed for VR, it will require controllers to play.

5.2 Software

- Virtual Vials will be downloaded as an .exe file and will mainly use the Unity VR engine.

6. References

- Elvira de Vries, L., & May, M. Virtual Laboratory Simulation in the Education of Laboratory Technicians - Motivation and Study Intensity. https://iubmb.onlinelibrary.wiley.com/doi/pdf/10.1002/bmb.21221
- Antonelli, D., Christopoulos, A., Laakso, M., Dagiene, V., Juskeviciene, A., Masiulionyte-Dagiene, V., Madziel, M., Stadnicka, D., Stylios, C. (2023). A Virtual Reality Laboratory for Blended Learning Education: Design, Implementation and Evaluation. https://doi.org/10.3390/educsci13050528